

A Model Monitoring Pipeline for the ASR Service, and How I Would Track Model Drift

HTX xData Technical Test — Task 8 (Essay)

Once the wav2vec2-large-960h service is in production, its accuracy is no longer guaranteed by offline benchmarks: the audio users submit will differ from LibriSpeech, and will keep changing. I would therefore deploy the model inside a monitoring pipeline with four stages: capture, aggregation, drift detection, and response.

Capture

The API is instrumented to log, for every request: input features (duration, sample rate, loudness/SNR, and metadata such as accent, age and gender where available), output features (transcript length, vocabulary, words-per-second, and a confidence signal derived from the mean entropy of the CTC posteriors), and operational metrics (latency, error rates). Raw audio is retained only for a small consented sample to allow later labelling. Each log line carries the model version and preprocessing version, so any anomaly can be attributed to a specific release rather than to genuine drift in the traffic.

Aggregation

Operational metrics stream to Prometheus/Grafana for real-time dashboards and paging. Feature logs land in a warehouse table partitioned by day, alongside a frozen *reference window* — the feature distribution measured at deployment time (e.g. over cv-valid-dev), which serves as the baseline all future windows are compared against.

Drift detection

A scheduled daily job compares the most recent window against the reference, separating three kinds of drift. *Data (covariate) drift* — the inputs change: detected with Population Stability Index or KL divergence on categorical features (accent, gender) and Kolmogorov–Smirnov tests on continuous ones (duration, loudness). *Prediction drift* — the outputs change: shifts in transcript length, out-of-vocabulary rate, empty-transcription rate, and especially a falling confidence distribution, which is the best label-free early warning of degraded accuracy. *Concept drift* — the input–output relationship itself changes: measured directly as word error rate on a small, continuously labelled sample (roughly 50–100 clips per week transcribed by humans), stratified by accent and gender so that degradation affecting a subgroup is not averaged away.

Response

Each metric has explicit thresholds (e.g. PSI > 0.2, WER more than three standard deviations above baseline) that trigger alerts. The playbook is tiered: investigate and slice the drifted segment; if

accuracy has genuinely degraded, fine-tune or retrain on recent labelled data; validate the candidate model in shadow mode against live traffic; then promote it via a canary rollout with automatic rollback. Every retrained model, its training window and its evaluation results are versioned in a model registry, closing the loop: the new model's distributions become the next reference window.

Two failure modes deserve special care. Sudden drift (a new client starts uploading 8 kHz telephony audio) is caught within hours by the input-distribution monitors; gradual drift (slang and named entities evolving over months) shows up first in the labelled-sample WER trend, which is why the human-labelled stream must never be paused, however small.

This design keeps the expensive part — human labelling — small, while the cheap, automated signals (input statistics and confidence scores) run continuously, so drift is caught in days rather than after users complain.

Word count: ≈500